HASEEB

VU

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M.SC

(MATH)

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Narowal Campus

MTH104 - Sets and Logic (Quiz # 1) Quiz

Question # 1 of 10 (Start time: 05:51:15 AM, 11 December 2021)

Let a and b be real numbers. Then $|a-b| \leq$ _____.

Select the correct option

- ● $|a|-|b|$
- ○ $|a|+|b|$
- ○ none

A set that contains the elements present in the universal set but not in set A is called the complement of set A .

KINEMASTER

Select the correct option

Reload Math

- True
- False

Sav

Question # 3 of 10 (Start time: 05:52:37 AM, 11 December 2021)

Suppose A and B are finite sets. Then $n(A \setminus B)=n(A) - n(A \cap B)$.

KINEMASTER

Select the correct option

- Given statement is false.
- Given statement is true.

MTH104 - Sets and Logic (Quiz # 1) Quiz Start Time: 05:51 AM, 11 December 2021

Question # 4 of 10 (Start time: 05:54:09 AM, 11 December 2021) Total Marks: 1

Sets are generally represented by small letters.

Select the correct option

Reload Math Equations

- False
- True

Saving

MTH104 - Sets and Logic (Quiz # 1) Quiz Start Time: 05:51 AM, 11 December 2021

Question # 5 of 10 (Start time: 05:54:21 AM, 11 December 2021) Total Marks: 1

Let a and b be real numbers. Geometrically, $a < b$ if and only if _____.

Select the correct option

Reload Math Equations

●	the point a lies to the left of the point b
○	the point a lies to the right of the point b

Saving...

MTH104 - Sets and Logic (Quiz # 1)

Quiz Start Time: 05:51 AM, 11 December 2021

Question # 6 of 10 (Start time: 05:54:44 AM, 11 December 2021)

Total Marks: 1

An odd element in the set $\{2, 4, 6, 7\}$ is _____.

Select the correct option

Reload Math Equations

○	2
○	4
○	6
●	7

MTH104 - Sets and Logic (Quiz # 1)

Quiz Start Time: 05:51 AM, 11 December 2021

Question # 7 of 10 (Start time: 05:55:05 AM, 11 December 2021)

Total Marks: 1

Which of the following set is bounded?

Select the correct option

Reload Math Equations

- $A = \{x : x > 2\}$
- none
- $C = \{3,4,5\}$
- $B = \{x : x < 3\}$

MTH104 - Sets and Logic (Quiz # 1)

Quiz Start Time: 05:51 AM, 11 December 2021

Question # 8 of 10 (Start time: 05:55:25 AM, 11 December 2021)

Total Marks: 1

The interval satisfying the inequality $|x| < -3$ is _____.

Select the correct option

Reload Math Equations

- none
- (-3,3)
- [-3,3]
- [-3,3)

MTH104 - Sets and Logic (Quiz # 1)

Quiz Start Time: 05:51 AM, 11 December 2021

Question # 9 of 10 (Start time: 05:55:53 AM, 11 December 2021)

Total Marks: 1

The interval satisfying the inequality $0 \leq x+2 < 3$ is _____.

Select the correct option

Reload Math Equations

○	none
○	[-2,1)
●	(-2,1)
○	[-2,1]

MTH104 - Sets and Logic (Quiz # 1) Quiz Start Time: 05:51 AM, 11 December 2021

Question # 10 of 10 (Start time: 05:57:08 AM, 11 December 2021) Total Marks: 1

gcd(31, 15) = _____.

Select the correct option

Reload Math Equations

○	none
●	1
○	15
○	31

Saving ...

MTH104 - Sets and Logic (Quiz # 1) Quiz Start Time: 10:52 AM, 11 December 2021

Question # 1 of 10 (Start time: 10:52:14 AM, 11 December 2021) **Total Marks:** 1

The interval satisfying the inequality $0 \leq 2x \leq 3$ is _____.

Select the correct option

Reload Math Equations

- (0,1/3)
- (0, 1/3]
- none
- [0,1/3]

Click to Save Answer & Move to Next Question

MTH104 - Sets and Logic (Quiz # 1)

Quiz Start Time: 10:52 AM, 11 December 2021

Question # 2 of 10 (Start time: 10:53:35 AM, 11 December 2021)

Total Marks: 1

The distance between -2 and 14 on the real line is _____.

Select the correct option

Reload Math Equations

- (●) 16
- ( ) 15
- ( ) -16
- ( ) 14

Click to Save Answer & Move to Next Question

MTH104 - Sets and Logic (Quiz # 1) Quiz Start Time: 10:52 AM, 11 December 2021

Question # 3 of 10 (Start time: 10:54:46 AM, 11 December 2021) **Total Marks:** 1

Number of elements in power set of a set {a,b,c} is _____.

Select the correct option

Reload Math Equations

○	3
○	none of these
●	8
○	2

Click to Save Answer & Move to Next Question

MTH104 - Sets and Logic (Quiz # 1) Quiz Start Time: 10:52 AM, 11 December 2021

Question # 4 of 10 (Start time: 10:55:28 AM, 11 December 2021) **Total Marks:** 1

Let a and b be real numbers. Then $|a \pm b| \geq ||a|| - ||b||$.

Select the correct option

Reload Math Equations

- ○ False
- ◉ True

Click to Save Answer & Move to Next Question

MTH104 - Sets and Logic (Quiz # 1)

Quiz Start Time: 10:52 AM, 11 December 2021

Question # 5 of 10 (Start time: 10:57:11 AM, 11 December 2021)

Total Marks: 1

_____ is "Two sets A and B are equal if and only if they have the same elements."

Select the correct option

Reload Math Equations

- (●) Principle of Extension
- ( ) Principle of Abstraction

Click to Save Answer & Move to Next Question

MTH104 - Sets and Logic (Quiz # 1) **Quiz Start Time: 10:52 AM, 11 December 2021**

Question # 6 of 10 (Start time: 10:57:53 AM, 11 December 2021) **Total Marks:** 1

Commutative laws are $A \cup B = B \cap A$ and $A \cap B = B \cup A$.

Select the correct option

Reload Math Equations

- ○ Given statement is true.
- ◉ Given statement is false.

Click to Save Answer & Move to Next Question

MTH104 - Sets and Logic (Quiz # 1)

Quiz Start Time: 10:52 AM, 11 December 2021

Question # 7 of 10 (Start time: 10:58:54 AM, 11 December 2021)

Total Marks: 1

The interval satisfying the inequality $|x| < -3$ is _____.

Select the correct option

Reload Math Equations

- ○ none
- ◉ (-3,3)
- ○ [-3,3]
- ○ [-3,3)

Click to Save Answer & Move to Next Question

MTH104 - Sets and Logic (Quiz # 1) Quiz Start Time: 10:52 AM, 11 December 2021

Question # 8 of 10 (Start time: 11:00:17 AM, 11 December 2021) **Total Marks:** 1

Let A be a set of real numbers. If for every $x \in A$, $|x| \leq M$, where M is a real number, then _____.

Select the correct option

Reload Math Equations

- (●) A is bounded
- ( ) A is unbounded

Click to Save Answer & Move to Next Question

MTH104 - Sets and Logic (Quiz # 1) Quiz Start Time: 10:52 AM, 11 December 2021

Question # 9 of 10 (Start time: 11:01:40 AM, 11 December 2021) Total Marks: 1

If A = {1,2,3} and B = {3,4,5} then A ∩ B is _____.

Select the correct option

Reload Math Equations

- ○ none of these
- ○ {3,4,5}
- ○ {1,2,3}
- ◉ {3}

Saving...

MTH104 - Sets and Logic (Quiz # 1)

Quiz Start Time: 10:52 AM, 11 December 2021

Question # 10 of 10 (Start time: 11:01:48 AM, 11 December 2021)

Total Marks: 1

Sets are generally represented by small letters.

Select the correct option

Reload Math Equations

- ○ True
- ◉ False

Click to Save Answer & Move to Next Question

MTH104 - Sets and Logic (Quiz # 1)

Quiz Start Time: 11:03 AM, 11 December 2021

Question # 1 of 10 (Start time: 11:03:01 AM, 11 December 2021)

Total Marks: 1

Number of elements in power set of a set {a,b,c} is _____.

Select the correct option

Reload Math Equations

- (●) 8
- ( ) none of these
- ( ) 2
- ( ) 3

Saving...

MTH104 - Sets and Logic (Quiz # 1)

Quiz Start Time: 11:03 AM, 11 December 2021

Question # 2 of 10 (Start time: 11:03:08 AM, 11 December 2021)

Total Marks: 1

Number of elements in power set of a set {1,2,3,4} is _____.

Select the correct option

Reload Math Equations

○	none of these
◉	16
○	8
○	4

Click to Save Answer & Move to Next Question

MTH104 - Sets and Logic (Quiz # 1)

Quiz Start Time: 11:03 AM, 11 December 2021

Question # 3 of 10 (Start time: 11:03:17 AM, 11 December 2021)

Total Marks: 1

The set Q of rational numbers is complete.

Select the correct option

Reload Math Equations

- (●) True
- ( ) False

Click to Save Answer & Move to Next Question

MTH104 - Sets and Logic (Quiz # 1)

Quiz Start Time: 11:03 AM, 11 December 2021

Question # 4 of 10 (Start time: 11:03:49 AM, 11 December 2021)

Total Marks: 1

Commutative laws are AUB=B∩A and A∩B=BUA.

Select the correct option

Reload Math Equations

- (●) Given statement is false.
- ( ) Given statement is true.

Saving...

MTH104 - Sets and Logic (Quiz # 1)

Quiz Start Time: 11:03 AM, 11 December 2021

Question # 5 of 10 (Start time: 11:03:57 AM, 11 December 2021)

Total Marks: 1

Sets are generally represented by small letters.

Select the correct option

Reload Math Equations

- (●) False
- ( ) True

Click to Save Answer & Move to Next Question

MTH104 - Sets and Logic (Quiz # 1)

Quiz Start Time: 11:03 AM, 11 December 2021

Question # 6 of 10 (Start time: 11:04:03 AM, 11 December 2021)

Total Marks: 1

The interval satisfying the inequality $0 \leq x+2 < 3$ is _____.

Select the correct option

Reload Math Equations

○	none
●	[-2,1)
○	(-2,1)
○	[-2,1]

Saving...

MTH104 - Sets and Logic (Quiz # 1) Quiz Start Time: 11:03 AM, 11 December 2021

Question # 7 of 10 (Start time: 11:04:20 AM, 11 December 2021) **Total Marks:** 1

If A = {1,2,3} and B = {3,4,5} then A ∩ B is _____.

Select the correct option

Reload Math Equations

○	none of these
●	{3}
○	{3,4,5}
○	{1,2,3}

Saving...

MTH104 - Sets and Logic (Quiz # 1)

Quiz Start Time: 11:03 AM, 11 December 2021

Question # 8 of 10 (Start time: 11:04:25 AM, 11 December 2021)

Total Marks: 1

The distance between 10 and 14 on the real line is _____.

Select the correct option

Reload Math Equations

- ○ 3
- ● 4
- ○ 0
- ○ -4

Click to Save Answer & Move to Next Question

MTH104 - Sets and Logic (Quiz # 1)

Quiz Start Time: 11:03 AM, 11 December 2021

Question # 9 of 10 (Start time: 11:04:38 AM, 11 December 2021)

Total Marks: 1

gcd(31, 15) = _____.

Select the correct option

Reload Math Equations

- (●) 1
- ( ) 31
- ( ) 15
- ( ) none

Click to Save Answer & Move to Next Question

MTH104 - Sets and Logic (Quiz # 1) Quiz Start Time: 11:03 AM, 11 December 2021

Question # 10 of 10 (Start time: 11:04:49 AM, 11 December 2021) **Total Marks:** 1

An odd element in the set {2, 4, 6, 7} is _____.

Select the correct option

Reload Math Equations

- ○ 4
- ○ 2
- ○ 6
- ◉ 7

Click to Save Answer & Move to Next Question

MTH104 - Sets and Logic (Quiz # 1)

Quiz Start Time: 11:06 AM, 11 December 2021

Question # 1 of 10 (Start time: 11:06:18 AM, 11 December 2021)

Total Marks: 1

The interval satisfying the inequality $0 \leq 2x \leq 3$ is _____.

Select the correct option

Reload Math Equations

- [ ] [0,1/3]
- [ ] (0, 1/3]
- [x] none
- [ ] (0,1/3)

Saving...

MTH104 - Sets and Logic (Quiz # 1)

Quiz Start Time: 11:06 AM, 11 December 2021

Question # 2 of 10 (Start time: 11:06:28 AM, 11 December 2021)

Total Marks: 1

The set R of real numbers is complete.

Select the correct option

Reload Math Equations

- ○ True
- ◉ False

Click to Save Answer & Move to Next Question

MTH104 - Sets and Logic (Quiz # 1) Quiz Start Time: 11:06 AM, 11 December 2021

Question # 3 of 10 (Start time: 11:06:57 AM, 11 December 2021) **Total Marks:** 1

The interval satisfying the inequality $|x| < -3$ is _____.

Select the correct option

Reload Math Equations

- none
- [-3,3)
- (-3,3)
- [-3,3]

Saving...

MTH104 - Sets and Logic (Quiz # 1)

Quiz Start Time: 11:06 AM, 11 December 2021

Question # 4 of 10 (Start time: 11:07:03 AM, 11 December 2021)

Total Marks: 1

An odd element in the set {2, 4, 6, 7} is _____.

Select the correct option

Reload Math Equations

- ○ 4
- ○ 6
- ◉ 7
- ○ 2

Click to Save Answer & Move to Next Question

MTH104 - Sets and Logic (Quiz # 1) Quiz Start Time: 11:06 AM, 11 December 2021

Question # 5 of 10 (Start time: 11:07:12 AM, 11 December 2021) **Total Marks:** 1

Identity laws are _____.

Select the correct option

Reload Math Equations

- ○ AU∅=A, A∩U=A
- ● both AU∅=A, A∩U=A and AUU=U, A∩∅=∅
- ○ AUU=U, A∩∅=∅
- ○ none of these

Click to Save Answer & Move to Next Question

MTH104 - Sets and Logic (Quiz # 1) Quiz Start Time: 11:06 AM, 11 December 2021

Question # 6 of 10 (Start time: 11:07:39 AM, 11 December 2021) Total Marks: 1

If a and b are positive, _____.

Select the correct option

Reload Math Equations

- then $a+b$ is positive and ab is negative.
- then $a+b$ is negative and ab is negative.
- then $a+b$ is positive and ab is positive.
- then $a+b$ is negative and ab is positive.

Saving...

MTH104 - Sets and Logic (Quiz # 1)

Quiz Start Time: 11:06 AM, 11 December 2021

Question # 7 of 10 (Start time: 11:07:51 AM, 11 December 2021)

Total Marks: 1

Real numbers cannot be represented as a point on a straight line.

Select the correct option

Reload Math Equations

- ○ True
- ◉ False

Saving...

MTH104 - Sets and Logic (Quiz # 1) Quiz Start Time: 11:06 AM, 11 December 2021

Question # 8 of 10 (Start time: 11:08:17 AM, 11 December 2021) **Total Marks:** 1

_____ is "Two sets A and B are equal if and only if they have the same elements."

Select the correct option

Reload Math Equations

- ○ Principle of Abstraction
- ◉ Principle of Extension

Saving...

MTH104 - Sets and Logic (Quiz # 1)

Quiz Start Time: 11:06 AM, 11 December 2021

Question # 9 of 10 (Start time: 11:08:23 AM, 11 December 2021)

Total Marks: 1

A set that contains the elements present in the universal set but not in set A is called the complement of set A .

Select the correct option

Reload Math Equations

- ○ False
- ● True

Click to Save Answer & Move to Next Question

MTH104 - Sets and Logic (Quiz # 1)

Quiz Start Time: 11:06 AM, 11 December 2021

Question # 10 of 10 (Start time: 11:08:54 AM, 11 December 2021)

Total Marks: 1

If two sets A and B are given, then the set consisting of all the elements of A and B is called Intersection of A and B.

Select the correct option

Reload Math Equations

- ○ True
- ◉ False

Click to Save Answer & Move to Next Question

MTH104 - Sets and Logic (Quiz # 1)

Quiz Start Time: 11:10 AM, 11 December 2021

Question # 1 of 10 (Start time: 11:10:17 AM, 11 December 2021)

Total Marks: 1

Number of elements in power set of a set {a,b,c} is _____.

Select the correct option

Reload Math Equations

- 2
- none of these
- 8
- 3

Saving...

MTH104 - Sets and Logic (Quiz # 1) Quiz Start Time: 11:10 AM, 11 December 2021

Question # 2 of 10 (Start time: 11:10:23 AM, 11 December 2021) **Total Marks:** 1

Those real numbers which cannot be represented as the ratio of integers, are called _____.

Select the correct option

Reload Math Equations

- (●) irrational numbers
- ( ) rational numbers

Saving...

MTH104 - Sets and Logic (Quiz # 1) Quiz Start Time: 11:10 AM, 11 December 2021

Question # 3 of 10 (Start time: 11:10:33 AM, 11 December 2021) **Total Marks:** 1

Let *a* and *b* be real numbers. Then $|a \pm b| \geq ||a|| - ||b||$.

Select the correct option

Reload Math Equations

- ○ False
- ● True

Saving...

MTH104 - Sets and Logic (Quiz # 1) Quiz Start Time: 11:10 AM, 11 December 2021

Question # 4 of 10 (Start time: 11:10:38 AM, 11 December 2021) **Total Marks:** 1

Let a and b be real numbers. Then $|a-b| \leq$ _____.

Select the correct option

Reload Math Equations

- ○ $|a|-|b|$
- ○ $|a||b|$
- ○ none
- ◉ $|a|+|b|$

Click to Save Answer & Move to Next Question

MTH104 - Sets and Logic (Quiz # 1)

Quiz Start Time: 11:10 AM, 11 December 2021

Question # 5 of 10 (Start time: 11:12:24 AM, 11 December 2021)

Total Marks: 1

Cardinality of set {{a,b}} is _____.

Select the correct option

Reload Math Equations

○	1
●	2
○	none of these
○	0

Click to Save Answer & Move to Next Question

MTH104 - Sets and Logic (Quiz # 1)

Quiz Start Time: 11:10 AM, 11 December 2021

Question # 6 of 10 (Start time: 11:13:31 AM, 11 December 2021)

Total Marks: 1

Which of the following set is unbounded?

Select the correct option

Reload Math Equations

- ○ $A = \{x : |x| < 4\}$
- ○ $B = \{x : |x| \leq 3\}$
- ◉ $C = \{1,2,3,...\}$
- ○ none

Saving...

MTH104 - Sets and Logic (Quiz # 1)

Quiz Start Time: 11:10 AM, 11 December 2021

Question # 7 of 10 (Start time: 11:13:44 AM, 11 December 2021)

Total Marks: 1

If A = {1,2,3} and B = {3,4,5} then A ∪ B is _____.

Select the correct option

Reload Math Equations

- ○ {1,2,3,4}
- ◉ none of these
- ○ {3,4,5}
- ○ {1,2,3}

Click to Save Answer & Move to Next Question

MTH104 - Sets and Logic (Quiz # 1)

Quiz Start Time: 11:10 AM, 11 December 2021

Question # 8 of 10 (Start time: 11:13:57 AM, 11 December 2021)

Total Marks: 1

Idempotent laws are $A \cup A = A$ and $A \cap A = A$.

Select the correct option

Reload Math Equations

- ○ False
- ● True

Click to Save Answer & Move to Next Question

MTH104 - Sets and Logic (Quiz # 1) Quiz Start Time: 11:10 AM, 11 December 2021

Question # 9 of 10 (Start time: 11:14:15 AM, 11 December 2021) Total Marks: 1

If p is a prime number and p does not divide a, then gcd(p, a) = _____.

Select the correct option

Reload Math Equations

- (●) 1
- ( ) none
- ( ) a
- ( ) p

Saving...

MTH104 - Sets and Logic (Quiz # 1) **Quiz Start Time: 11:10 AM, 11 December 2021**

Question # 10 of 10 (Start time: 11:14:33 AM, 11 December 2021) **Total Marks:** 1

Suppose A and B are finite sets. Then $n(A \setminus B)=n(A) - n(A \cap B)$.

Select the correct option

Reload Math Equations

○	Given statement is false.
●	Given statement is true.

Click to Save Answer & Move to Next Question

MTH104 - Sets and Logic (Quiz # 1) **Quiz Start Time: 11:15 AM, 11 December 2021**

Question # 1 of 10 (Start time: 11:15:59 AM, 11 December 2021) **Total Marks:** 1

Which of the following set is unbounded?

Select the correct option

Reload Math Equations

○	none
●	C = {1,2,3,...}
○	B = {x : |x| ≤ 3}
○	A = {x : |x| < 4}

Saving...

MTH104 - Sets and Logic (Quiz # 1)

Quiz Start Time: 11:15 AM, 11 December 2021

Question # 2 of 10 (Start time: 11:16:04 AM, 11 December 2021)

Total Marks: 1

The set R of real numbers is complete.

Select the correct option

Reload Math Equations

- (•) False
- ( ) True

Click to Save Answer & Move to Next Question

MTH104 - Sets and Logic (Quiz # 1)

Quiz Start Time: 11:15 AM, 11 December 2021

Question # 3 of 10 (Start time: 11:16:13 AM, 11 December 2021)

Total Marks: 1

Real numbers cannot be represented as a point on a straight line.

Select the correct option

Reload Math Equations

- (●) False
- ( ) True

Saving...

MTH104 - Sets and Logic (Quiz # 1)

Quiz Start Time: 11:15 AM, 11 December 2021

Question # 4 of 10 (Start time: 11:16:18 AM, 11 December 2021)

Total Marks: 1

If p is a prime number and p does not divide a, then gcd(p, a) = _____.

Select the correct option

Reload Math Equations

○	a
○	none
●	1
○	p

Saving...

MTH104 - Sets and Logic (Quiz # 1)

Quiz Start Time: 11:15 AM, 11 December 2021

Question # 5 of 10 (Start time: 11:16:24 AM, 11 December 2021)

Total Marks: 1

If A = {1,2,3} and B = {3,4,5} then A ∩ B is _____.

Select the correct option

Reload Math Equations

- (•) {3}
- ( ) {1,2,3}
- ( ) none of these
- ( ) {3,4,5}

Saving...

MTH104 - Sets and Logic (Quiz # 1)

Quiz Start Time: 11:15 AM, 11 December 2021

Question # 6 of 10 (Start time: 11:16:30 AM, 11 December 2021)

Total Marks: 1

Two distinct points are on one and only one line.

Select the correct option

Reload Math Equations

- (●) True
- ( ) False

Click to Save Answer & Move to Next Question

MTH104 - Sets and Logic (Quiz # 1)

Quiz Start Time: 11:15 AM, 11 December 2021

Question # 7 of 10 (Start time: 11:16:56 AM, 11 December 2021)

Total Marks: 1

The interval satisfying the inequality $0 \leq x+2 < 3$ is _____.

Select the correct option

Reload Math Equations

- ○ none
- ◉ [-2,1)
- ○ [-2,1]
- ○ (-2,1)

Saving...

MTH104 - Sets and Logic (Quiz # 1) Quiz Start Time: 11:15 AM, 11 December 2021

Question # 8 of 10 (Start time: 11:17:06 AM, 11 December 2021) **Total Marks:** 1

Let A be a set of real numbers. If for every $x \in A$, $|x| \leq M$, where M is a real number, then _____.

Select the correct option

Reload Math Equations

- ○ A is unbounded
- ◉ A is bounded

Click to Save Answer & Move to Next Question

MTH104 - Sets and Logic (Quiz # 1)

Quiz Start Time: 11:15 AM, 11 December 2021

Question # 9 of 10 (Start time: 11:17:18 AM, 11 December 2021)

Total Marks: 1

The set Q of rational numbers is complete.

Select the correct option

Reload Math Equations

- (●) True
- ( ) False

Click to Save Answer & Move to Next Question

MTH104 - Sets and Logic (Quiz # 1)

Quiz Start Time: 11:15 AM, 11 December 2021

Question # 10 of 10 (Start time: 11:17:29 AM, 11 December 2021)

Total Marks: 1

Two distinct lines cannot contain more than one point in common.

Select the correct option

Reload Math Equations

- (●) Given statement is false.
- ( ) Given statement is true.

Click to Save Answer & Move to Next Question

MTH104 - Sets and Logic (Quiz # 1)

Quiz Start Time: 11:18 AM, 11 December 2021

Question # 1 of 10 (Start time: 11:18:27 AM, 11 December 2021)

Total Marks: 1

A group or collection of well-defined objects is called a set.

Select the correct option

Reload Math Equations

- ○ False
- ● True

Click to Save Answer & Move to Next Question

MTH104 - Sets and Logic (Quiz # 1) Quiz Start Time: 11:18 AM, 11 December 2021

Question # 2 of 10 (Start time: 11:18:45 AM, 11 December 2021) **Total Marks:** 1

Cardinality of set {{a,b}} is _____.

Select the correct option

Reload Math Equations

○	1
○	0
●	2
○	none of these

Saving...

MTH104 - Sets and Logic (Quiz # 1) Quiz Start Time: 11:18 AM, 11 December 2021

Question # 3 of 10 (Start time: 11:18:51 AM, 11 December 2021) **Total Marks:** 1

Let A be a set of real numbers. If for every $x \in A$, $|x| \leq M$, where M is a real number, then _____.

Select the correct option

Reload Math Equations

- (●) A is bounded
- ( ) A is unbounded

Saving...

MTH104 - Sets and Logic (Quiz # 1)

Quiz Start Time: 11:18 AM, 11 December 2021

Question # 4 of 10 (Start time: 11:18:56 AM, 11 December 2021)

Total Marks: 1

The distance between -2 and 14 on the real line is _____.

Select the correct option

Reload Math Equations

- -16
- 14
- 16
- 15

Click to Save Answer & Move to Next Question

MTH104 - Sets and Logic (Quiz # 1)

Quiz Start Time: 11:18 AM, 11 December 2021

Question # 5 of 10 (Start time: 11:19:01 AM, 11 December 2021)

Total Marks: 1

If two sets A and B are given, then the set consisting of all the elements of A and B is called Intersection of A and B.

Select the correct option

Reload Math Equations

- (●) False
- ( ) True

Saving...

MTH104 - Sets and Logic (Quiz # 1)

Quiz Start Time: 11:18 AM, 11 December 2021

Question # 6 of 10 (Start time: 11:19:09 AM, 11 December 2021)

Total Marks: 1

The interval satisfying the inequality $|x| < -3$ is _____.

Select the correct option

Reload Math Equations

- [ ] [-3,3)
- [ ] none
- [x] (-3,3)
- [ ] [-3,3]

Click to Save Answer & Move to Next Question

MTH104 - Sets and Logic (Quiz # 1) Quiz Start Time: 11:18 AM, 11 December 2021

Question # 7 of 10 (Start time: 11:19:26 AM, 11 December 2021) **Total Marks:** 1

If p is a prime number and p | a, then gcd(p, a) = _____.

Select the correct option

Reload Math Equations

- ○ p
- ○ a
- ○ a+p
- ◉ none

Click to Save Answer & Move to Next Question

MTH104 - Sets and Logic (Quiz # 1)

Quiz Start Time: 11:18 AM, 11 December 2021

Question # 8 of 10 (Start time: 11:19:40 AM, 11 December 2021)

Total Marks: 1

If A = {1,2,3} and B = {3,4,5} then A ∩ B is _____.

Select the correct option

Reload Math Equations

- (•) {3}
- ( ) none of these
- ( ) {3,4,5}
- ( ) {1,2,3}

Saving...

MTH104 - Sets and Logic (Quiz # 1)

Quiz Start Time: 11:18 AM, 11 December 2021

Question # 9 of 10 (Start time: 11:19:47 AM, 11 December 2021)

Total Marks: 1

Those real numbers which cannot be represented as the ratio of integers, are called _____.

Select the correct option

Reload Math Equations

- irrational numbers
- rational numbers

Click to Save Answer & Move to Next Question

MTH104 - Sets and Logic (Quiz # 1) Quiz Start Time: 11:18 AM, 11 December 2021

Question # 10 of 10 (Start time: 11:19:55 AM, 11 December 2021) **Total Marks:** 1

Which of the following set is bounded?

Select the correct option

Reload Math Equations

- ○ $B = \{x : x < 3\}$
- ○ $A = \{x : x > 2\}$
- ○ none
- ◉ $C = \{3,4,5\}$

Click to Save Answer & Move to Next Question

MTH104 - Sets and Logic (Quiz # 1)

Quiz Start Time: 11:21 AM, 11 December 2021

Question # 1 of 10 (Start time: 11:21:40 AM, 11 December 2021) **Total Marks:** 1

Idempotent laws are $A \cup A = A$ and $A \cap A = A$.

Select the correct option

Reload Math Equations

- (●) True
- ( ) False

Saving...

MTH104 - Sets and Logic (Quiz # 1)

Quiz Start Time: 11:21 AM, 11 December 2021

Question # 2 of 10 (Start time: 11:21:45 AM, 11 December 2021)

Total Marks: 1

The set Q of rational numbers is complete.

Select the correct option

Reload Math Equations

- ○ False
- ◉ True

Saving...

MTH104 - Sets and Logic (Quiz # 1)

Quiz Start Time: 11:21 AM, 11 December 2021

Question # 3 of 10 (Start time: 11:21:50 AM, 11 December 2021)

Total Marks: 1

If A = {1,2,3} and B = {3,4,5} then A ∩ B is _____.

Select the correct option

Reload Math Equations

○	none of these
○	{1,2,3}
◉	{3}
○	{3,4,5}

Saving...

MTH104 - Sets and Logic (Quiz # 1)

Quiz Start Time: 11:21 AM, 11 December 2021

Question # 4 of 10 (Start time: 11:21:55 AM, 11 December 2021)

Total Marks: 1

Identity laws are _____.

Select the correct option

Reload Math Equations

- ○ $A \cup U = U$, $A \cap \emptyset = \emptyset$
- ○ $A \cup \emptyset = A$, $A \cap U = A$
- ◉ both $A \cup \emptyset = A$, $A \cap U = A$ and $A \cup U = U$, $A \cap \emptyset = \emptyset$
- ○ none of these

Saving...

MTH104 - Sets and Logic (Quiz # 1)

Quiz Start Time: 11:21 AM, 11 December 2021

Question # 5 of 10 (Start time: 11:22:00 AM, 11 December 2021)

Total Marks: 1

Suppose A and B are finite sets. Then $n(A \setminus B)=n(A) - n(A \cap B)$.

Select the correct option

Reload Math Equations

- (●) Given statement is true.
- ( ) Given statement is false.

Saving...

MTH104 - Sets and Logic (Quiz # 1) Quiz Start Time: 11:21 AM, 11 December 2021

Question # 6 of 10 (Start time: 11:22:04 AM, 11 December 2021) **Total Marks:** 1

The interval satisfying the inequality $|x| < -3$ is _____.

Select the correct option

Reload Math Equations

- [ ] [-3,3)
- [ ] none
- [ ] [-3,3]
- [x] (-3,3)

Saving...

MTH104 - Sets and Logic (Quiz # 1) **Quiz Start Time: 11:21 AM, 11 December 2021**

Question # 7 of 10 (Start time: 11:22:08 AM, 11 December 2021) **Total Marks:** 1

Let a and b be real numbers. Geometrically, $a < b$ if and only if _____.

Select the correct option

Reload Math Equations

- (●) the point a lies to the left of the point b
- ( ) the point a lies to the right of the point b

Saving...

MTH104 - Sets and Logic (Quiz # 1) **Quiz Start Time: 11:21 AM, 11 December 2021**

Question # 8 of 10 (Start time: 11:22:58 AM, 11 December 2021) **Total Marks:** 1

The interval satisfying the inequality $0 \leq x+2 < 3$ is _____.

Select the correct option

Reload Math Equations

○	none
○	[-2,1]
●	[-2,1)
○	(-2,1)

Saving...

MTH104 - Sets and Logic (Quiz # 1)

Quiz Start Time: 11:21 AM, 11 December 2021

Question # 9 of 10 (Start time: 11:23:06 AM, 11 December 2021)

Total Marks: 1

A group or collection of well-defined objects is called a set.

Select the correct option

Reload Math Equations

- ○ False
- ● True

Click to Save Answer & Move to Next Question

MTH104 - Sets and Logic (Quiz # 1)

Quiz Start Time: 11:21 AM, 11 December 2021

Question # 10 of 10 (Start time: 11:23:12 AM, 11 December 2021)

Total Marks: 1

The distance between -2 and 14 on the real line is _____.

Select the correct option

Reload Math Equations

- 14
- 16
- 15
- -16

Click to Save Answer & Move to Next Question

MTH104 - Sets and Logic (Quiz # 1) Quiz Start Time: 11:24 AM, 11 December 2021

Question # 2 of 10 (Start time: 11:24:32 AM, 11 December 2021) **Total Marks:** 1

Commutative laws are $A \cup B = B \cap A$ and $A \cap B = B \cup A$.

Select the correct option

Reload Math Equations

- (●) Given statement is false.
- ( ) Given statement is true.

Saving...

MTH104 - Sets and Logic (Quiz # 1) Quiz Start Time: 11:24 AM, 11 December 2021

Question # 3 of 10 (Start time: 11:24:40 AM, 11 December 2021) **Total Marks:** 1

Cardinality of set {{a,b}} is _____.

Select the correct option

Reload Math Equations

- ○ none of these
- ◉ 2
- ○ 1
- ○ 0

Saving...

MTH104 - Sets and Logic (Quiz # 1)

Quiz Start Time: 11:24 AM, 11 December 2021

Question # 4 of 10 (Start time: 11:24:43 AM, 11 December 2021)

Total Marks: 1

If p is a prime number and p | a, then gcd(p, a) = _____.

Select the correct option

Reload Math Equations

- (●) none
- ( ) a
- ( ) a+p
- ( ) p

Saving...

MTH104 - Sets and Logic (Quiz # 1)

Quiz Start Time: 11:24 AM, 11 December 2021

Question # 5 of 10 (Start time: 11:24:47 AM, 11 December 2021)

Total Marks: 1

The set of integers Z is dense in R.

Select the correct option

Reload Math Equations

- (●) False
- ( ) True

Click to Save Answer & Move to Next Question

MTH104 - Sets and Logic (Quiz # 1)

Quiz Start Time: 11:24 AM, 11 December 2021

Question # 6 of 10 (Start time: 11:25:14 AM, 11 December 2021)

Total Marks: 1

The interval satisfying the inequality $0 \leq x+2 < 3$ is _____.

Select the correct option

Reload Math Equations

- ○ none
- ● [-2,1)
- ○ [-2,1]
- ○ (-2,1)

Click to Save Answer & Move to Next Question

MTH104 - Sets and Logic (Quiz # 1)

Quiz Start Time: 11:24 AM, 11 December 2021

Question # 7 of 10 (Start time: 11:25:21 AM, 11 December 2021)

Total Marks: 1

If A = {1,2,3} and B = {3,4,5} then A ∩ B is _____.

Select the correct option

Reload Math Equations

○	{1,2,3}
◉	{3}
○	none of these
○	{3,4,5}

Click to Save Answer & Move to Next Question

MTH104 - Sets and Logic (Quiz # 1)

Quiz Start Time: 11:24 AM, 11 December 2021

Question # 8 of 10 (Start time: 11:25:26 AM, 11 December 2021)

Total Marks: 1

Identity laws are _____.

Select the correct option

Reload Math Equations

○	none of these
●	both AU∅=A, A∩U=A and AUU=U, A∩∅=∅
○	AUU=U, A∩∅=∅
○	AU∅=A, A∩U=A

Saving...

MTH104 - Sets and Logic (Quiz # 1)

Quiz Start Time: 11:24 AM, 11 December 2021

Question # 9 of 10 (Start time: 11:25:31 AM, 11 December 2021)

Total Marks: 1

Suppose A and B are finite disjoint sets. Then $n(A \cup B)=n(A) - n(B)$.

Select the correct option

Reload Math Equations

- (●) Given statement is false.
- ( ) Given statement is true.

Click to Save Answer & Move to Next Question

MTH104 - Sets and Logic (Quiz # 1) Quiz Start Time: 11:24 AM, 11 December 2021

Question # 10 of 10 (Start time: 11:25:56 AM, 11 December 2021) **Total Marks:** 1

If A = {1,2,3} and B = {3,4,5} then A ∪ B is _____.

Select the correct option

Reload Math Equations

○	{1,2,3}
●	none of these
○	{1,2,3,4}
○	{3,4,5}

Click to Save Answer & Move to Next Question

MTH104 - Sets and Logic (Quiz # 1) **Quiz Start Time: 11:27 AM, 11 December 2021**

Question # 1 of 10 (Start time: 11:27:06 AM, 11 December 2021) **Total Marks: 1**

Let a and b be real numbers. Then $|a-b| \leq$ _____.

Select the correct option

Reload Math Equations

- ○ $|a|+|b|$
- ○ $|a||b|$
- ○ none
- ◉ $|a|-|b|$

Saving...

MTH104 - Sets and Logic (Quiz # 1)

Quiz Start Time: 11:27 AM, 11 December 2021

Question # 2 of 10 (Start time: 11:27:43 AM, 11 December 2021)

Total Marks: 1

Two distinct lines cannot contain more than one point in common.

Select the correct option

Reload Math Equations

- Given statement is true.
- Given statement is false.

Saving...

MTH104 - Sets and Logic (Quiz # 1)

Quiz Start Time: 11:27 AM, 11 December 2021

Question # 3 of 10 (Start time: 11:27:53 AM, 11 December 2021)

Total Marks: 1

The set R of real numbers is complete.

Select the correct option

Reload Math Equations

- (●) False
- ( ) True

Click to Save Answer & Move to Next Question

MTH104 - Sets and Logic (Quiz # 1)

Quiz Start Time: 11:27 AM, 11 December 2021

Question # 4 of 10 (Start time: 11:28:02 AM, 11 December 2021)

Total Marks: 1

Cardinality of empty set is 0.

Select the correct option

Reload Math Equations

- True
- False

Click to Save Answer & Move to Next Question

MTH104 - Sets and Logic (Quiz # 1) **Quiz Start Time: 11:27 AM, 11 December 2021**

Question # 5 of 10 (Start time: 11:28:35 AM, 11 December 2021) **Total Marks:** 1

Identity laws are _____.

Select the correct option

Reload Math Equations

○	AUU=U, A∩Ø=Ø
○	none of these
●	both AUØ=A, A∩U=A and AUU=U, A∩Ø=Ø
○	AUØ=A, A∩U=A

Click to Save Answer & Move to Next Question

MTH104 - Sets and Logic (Quiz # 1)

Quiz Start Time: 11:27 AM, 11 December 2021

Question # 6 of 10 (Start time: 11:28:49 AM, 11 December 2021)

Total Marks: 1

gcd(12, -18) = _____.

Select the correct option

Reload Math Equations

○	3
●	6
○	-3
○	-6

Saving...

MTH104 - Sets and Logic (Quiz # 1) Quiz Start Time: 11:27 AM, 11 December 2021

Question # 7 of 10 (Start time: 11:29:04 AM, 11 December 2021) **Total Marks:** 1

Cardinality of set {{a,b}} is _____.

Select the correct option

Reload Math Equations

○	0
○	1
●	2
○	none of these

Click to Save Answer & Move to Next Question

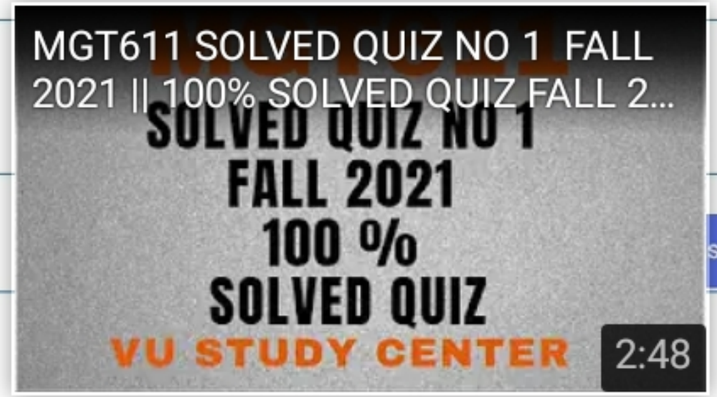
MTH104 - Sets and Logic (Quiz # 1)

Quiz Start Time: 11:27 AM, 11 December 2021

Question # 8 of 10 (Start time: 11:29:10 AM, 11 December 2021)

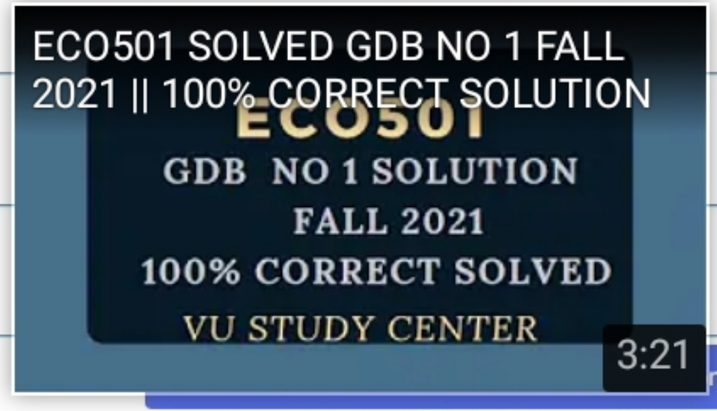
Total Marks: 1

Number of elements in power set of a set {1,2,3,4} is _____.



Select the correct option

- ○ none of these
- ○ 8
- ○ 4
- ◉ 16



MTH104 - Sets and Logic (Quiz # 1)

Quiz Start Time: 11:27 AM, 11 December 2021

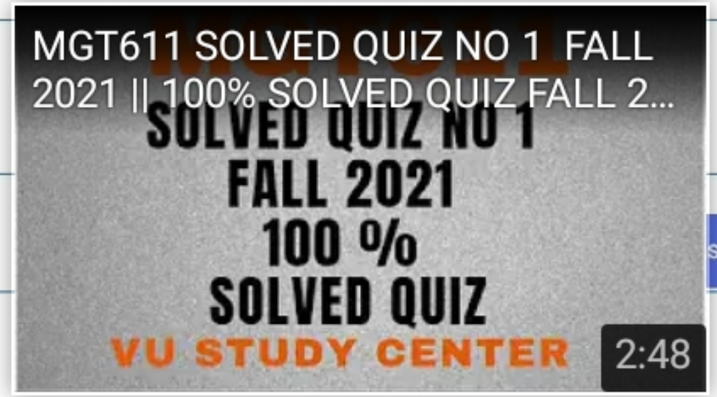
Question # 9 of 10 (Start time: 11:29:21 AM, 11 December 2021)

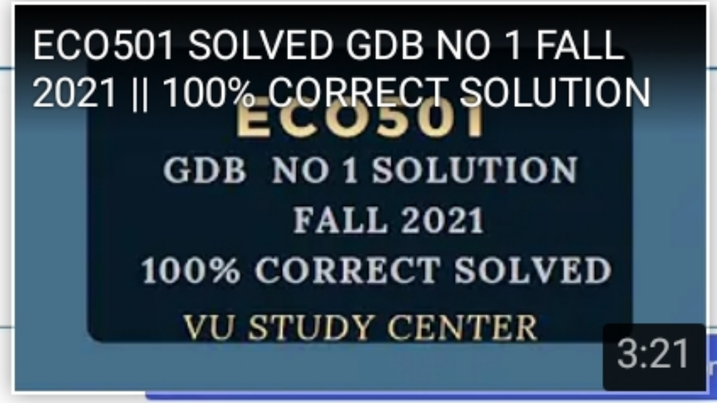
Total Marks: 1

Let *a* and *b* be real numbers. Then $|a \pm b| \geq ||a|| - ||b||$.

Select the correct option

- False
- True





MTH104 - Sets and Logic (Quiz # 1)

Quiz Start Time: 11:27 AM, 11 December 2021

Question # 10 of 10 (Start time: 11:29:26 AM, 11 December 2021)

Total Marks: 1

If *a* and *b* are positive, _____.

Select the correct option

- then *a+b* is negative and *ab* is positive.
- then *a+b* is positive and *ab* is negative.
- then *a+b* is positive and *ab* is positive.
- then *a+b* is negative and *ab* is negative.

